derivlab

A small, readable Python toolkit for pricing and risk-managing derivatives, built module by module along a 62-hour course on rates, corporate hedging, equity volatility and structured products. Every function is tested, every number in the course notebooks comes from executed code, and the core results match QuantLib to machine precision.

All market data is synthetic (plausible levels, no real market date and no bank data).

What it does

Area	What you can do
Rates	Build €STR and Euribor 6M curves from OIS, FRA and swap quotes (dual-curve bootstrapping); price FRAs, swaps, amortising and forward-starting swaps; bucketed DV01, PCA and hedging
Corporate hedging	Simulate Euribor with a Hull-White model fitted to the curves; interest cost at risk, efficient frontier of fixed/floating mixes; caps, floors, collars; inflation swaps; prehedging; xVA with and without CSA; cross-currency swaps
Rate options	Black, Bachelier and shifted Black; CMS and in-arrears convexity; caplet stripping and SABR; Hull-White calibration, trinomial tree, Bermudan and cancellable swaps
Equity	Index forwards with discrete dividends and repo; Black-Scholes, Greeks and binomial trees (American, lock-up, exercise boundary); delta-hedging P&L; SSVI surface, Dupire local vol and its Monte Carlo; Heston; variance swaps; dividend futures
Exotics and structured	Digitals and barriers (closed form, Broadie-Glasserman-Kou, Brownian bridge); worst-of, baskets, quantos and implied correlation; reverse convertibles, capital protection, Phoenix and Athena autocallables priced with local vol and issuer funding; book risk; corporate equity solutions

Repository layout

```
derivlab/      the library (pure Python: numpy, scipy, pandas)
tests/         80 pytest tests
notebooks/     22 course notebooks A01-E05, executed, each with a client case, checks and solved exercises
project/       final project: three demos and the QuantLib contrast
  P01_corporate_hedge.ipynb   curves -> exposure -> swap, cap, collar -> Hull-White cost scenarios -> recommendation
  P02_hedge_xva.ipynb         exposure, CVA/DVA/FVA of that swap, no CSA vs CSA vs CSA with threshold
  P03_autocall_risk.ipynb     Phoenix autocall: local-vol pricing, Greeks, risk ladder, stress-day P&L explain
  P04_quantlib_check.ipynb    curves, swaption, barriers and Hull-White bond options against QuantLib
exam/          numerical solutions of the 40-question final exam, with checks
INTERVIEW_GUIDE.md            one-page guide to present the project
```

Quick start

```
pip install -e ".[test]"          # library + pytest
pytest                            # 80 tests, about 20 seconds
pip install -e ".[notebooks]"     # to run the notebooks
```

```
jupyter lab notebooks/            # or project/
pip install -e ".[check]"         # QuantLib, only for project/P04
```

```
from derivlab.market import sample_market
from derivlab.curves import build_curves
from derivlab.dates import spot_date, add_months
from derivlab.swaps import par_swap_rate

m = sample_market()
estr, euribor6m = build_curves(m)
spot = spot_date(m["ref_date"])
print(par_swap_rate(m["ref_date"], spot, add_months(spot, 84), 1.0, estr, euribor6m))
# 7y swap, about 2.54%
```

Modules

Module	Part	Contents
`dates`	A	TARGET calendar, business-day conventions, day counts, schedules
`market`	A	Synthetic EUR market (€STR OIS, Euribor 6M fixing, FRAs, swaps)
`curves`	A	Dual-curve bootstrapping, log-linear discount factors
`swaps`	A	FRAs, swaps, amortising and forward-starting swaps, €STR/Euribor basis
`risk`	A	Bucketed DV01 by curve rebuild, PCA, least-squares hedging
`scenarios`	B	Hull-White (G1++) fitted to the curves, deterministic Euribor basis
`hedging`	B	Euribor scenarios, interest cost at risk, efficient frontier, strategy comparison
`rate_options`	B, C	Bachelier, Black, shifted Black; caps, floors, collars, swaptions; SABR; caplet stripping
`inflation`	B	Zero-coupon inflation swaps and real rates
`xva`	B	Simulated exposure, CVA, DVA, FCA, FBA, collateral and breaks
`fx`	B	SOFR curve, cross-currency basis, CCS, leverage-at-risk
`convexity`	C	CMS and in-arrears adjustments
`vol_market`	C	Synthetic swaption matrix, cap vols, 5y10y smile
`short_rate`	C	Vasicek; Hull-White bond options, swaptions, calibration, trinomial tree, Bermudans
`equity_market`	D	Synthetic index: spot, repo, dividend futures and forecasts, SSVI parameters
`equity`	D	Forwards, Black-Scholes and Greeks, CRR tree, delta hedging, SSVI, local vol MC, Heston, variance swaps
`exotics`	E	Digitals, call-spread replication, barriers, BGK, Brownian-bridge Monte Carlo
`multi_asset`	E	Correlated simulation, basket vol, implied correlation, quantos
`structured`	E	Index paths (local or flat vol), issuer funding, autocallables, reverse convertibles, capital protection
`corporate`	E	Collars, secured loans, total return swaps, prepaid forwards, conversion probability, dilution

Conventions

- Rates in decimals. Amounts from the holder's point of view (a payer pays fixed), except `xva`, which takes the bank's view.

- TARGET calendar, Modified Following, spot T+2. OIS: annual ACT/360. Euribor swaps: fixed annual 30/360 against 6M ACT/360. Curves: log-linear discount factors on ACT/365F time; €STR discounting for collateralised trades.
- Scenario work (Parts B and C) uses a time grid in years. Hull-White: mean reversion 0.03, normal vol 0.85% a year. "Real-world" scenarios subtract a term premium tp(t) = 0.60% (1 - exp(-t/3)).
- Equity: escrowed discrete dividends, Black-76 on the forward, implied vol as a function of k = ln(K/F). The local-vol Monte Carlo simulates ln(S/F) with log-Euler steps and antithetic variates.
- Structured notes are valued as the issuer's liability: cash flows discounted at €STR + funding spread; the issuer's margin is the issue price minus that value.

Validation

- 80 unit tests: textbook examples (Hull 30.1, 30.2), no-arbitrage identities (put-call parity, in-out parity, calendar and butterfly conditions), model limits (Heston with zero vol of vol is Black-Scholes) and Monte Carlo against closed forms.
- Every notebook ends with assertions; notebooks fail if a check fails.
- `project/P04_quantlib_check.ipynb` : €STR and Euribor 6M curves, a 5y10y Bachelier swaption, ten barrier options and six Hull-White bond options agree with QuantLib 1.44 to 1e-9 bp, 1e-4 EUR on 100 M and 1e-14 respectively.

Limitations

Synthetic data; flat-vol barrier and correlation analytics; local vol has unrealistic forward-smile dynamics (no stochastic-local vol); xVA without wrong-way risk and with an analytic margin-period approximation; no FX options, credit or commodities.